PART 4

Trauma, Burns and Environmental Emergencies

IN THIS PART

1. Use the mechanism of injury to raise suspicion of serious injury.
2. Describe the order of external bleeding control (pressure, packing, tourniquet) and the steps of each.
3. Recognize head, spine and chest injuries and state first priorities, including the principles of spinal motion restriction.
4. Describe basic splinting and the checks that go with it.
5. Estimate burn size and state first priorities.
6. Recognize heat and cold emergencies.

Injury is the emergency where what you see is often the least of it. A small wound may sit over a large bleed. A person who walks away from a crash may have an unstable spine. A burn that looks minor may cover a large share of the body. This Part teaches you to read the story of the injury before you decide what to do, and then to act on the problems that kill fastest.

The six entries move from the general to the particular. Entry 4.1 sets out how to size up a trauma patient and what the mechanism of injury tells you, and Entry 4.2 deals with bleeding, the one

trauma problem that a bystander can control in minutes. Injuries to the head, spine and chest come next, followed by broken bones and splinting. Burns have their own entry, and the Part closes with the emergencies caused by heat and cold.

Everything here builds on the safe scene and the primary survey of Part 1. The skills in this Part are basic and non-invasive, and each is taught in a course before it is done on a patient. Your scope of practice and local protocol decide what you may do.

4.1 Trauma Assessment and Mechanism of Injury

A trauma patient may look much better or much worse than they are. The mechanism of injury, meaning how the injury happened, is your best early clue to how worried to be.

WHAT IT IS. Trauma assessment applies the primary survey of Entry 1.3 to an injured patient, with two changes. First, as Entry 1.3 noted, life-threatening external bleeding is controlled before the airway, because a patient can lose a dangerous amount of blood in minutes; your protocol sets the exact order. Second, you consider the mechanism of injury from the first look and keep it in mind, so that the examination looks for the injuries that the mechanism makes likely.

WHY IT MATTERS. The body hides serious injury. A patient, and a child in particular, can keep a normal blood pressure for a time while bleeding inside, as Entry 3.5 explained for shock. A patient with a distracting injury, such as a broken leg, may not mention neck pain. If you wait for the patient to complain of every problem, you will miss some of them. The mechanism tells you where to look before the patient can.

HIGH-RISK MECHANISMS. Treat these as a warning that serious injury may be present even when the patient looks well: a fall from a height of more than about 10 feet (3 meters), a serious vehicle crash (for example, one in which the patient was thrown out, the passenger space was pushed in, or the patient had to be cut free), a rider thrown from a motorcycle or a similar vehicle, a crash in which another person in the same vehicle died, a pedestrian or cyclist thrown, run over or hit hard by a vehicle, a penetrating wound to the head, neck or trunk, and a crush injury. An injury that does not fit the story given, or repeated injuries in a child, may point to abuse, which Entry 6.6 covers. Age and condition change the risk. Older adults and people on blood-thinning medicines can bleed seriously after an apparently minor fall, and children may look well until they suddenly deteriorate.

> **REMEMBER**
>
> Read the scene before you read the patient: a smashed windshield, a bent steering wheel, a cracked helmet or a ladder beside the person tells you the mechanism before the patient can.

HIDDEN BLEEDING. Blood can collect where you cannot see it: in the chest, the abdomen, the pelvis and around a broken thigh bone. A patient with a tender, swollen or rigid abdomen, a painful pelvis, or a thigh that is swollen and deformed may be losing a great deal of blood inside the body. Do not press on the pelvis to test it, and do not rock it. Leave the examination of these areas to the clinicians, report what you see, and treat the signs of shock as they appear.

THE TRAUMA SURVEY. Begin with a safe scene and a look for bleeding. Control severe bleeding at once. Then work through airway, breathing, circulation, disability and exposure, using the jaw thrust in preference to the head tilt if a spinal injury is possible, as

Entry 2.2 described. Expose the body enough to look for injuries and cover the patient again. Take vital signs and write down the times. Call for help early, since a patient with a significant mechanism and any sign of instability needs a trauma center, and transport decisions belong to the responding clinicians.

RED FLAGS

Severe bleeding, a patient who is confused or drowsy after an injury, difficulty breathing, a rigid or swollen abdomen, a deformed or unstable pelvis, a fast weak pulse with pale cool skin, or a high-risk mechanism with any of these. Call for help at once.

KEEPING THE PATIENT WARM. An injured patient loses heat fast, especially when wet or lying on the ground, and a cold patient may clot blood less well. Cover the patient with a blanket or coat, put something between the patient and the ground, and remove wet clothing where you can do so without moving an injured limb. Do this while you assess, because the habit protects the patient from the start. Take care not to overheat a patient in a hot setting.

REASSESSING AFTER INJURY. The first findings in an injured patient are only the first. Repeat the primary survey at short intervals and after any move. Recheck bleeding, since a dressing that was dry may be wet, and recheck the level of consciousness, since a patient with a head injury can worsen after an alert interval. Record each set of findings with the time and tell the team that takes over what changed.

NURSING AND RESPONDER PRIORITIES. Control severe bleeding first, then work through the primary survey with the spine in mind, and call early for any high-risk mechanism.

See also: Entry 1.3 (primary survey); Entry 3.5 (shock); Entry 4.2 (bleeding); Entry 4.3 (spine); Entry 6.1 (children); Entry 6.6 (abuse).

4.2 Bleeding and Hemorrhage Control

Severe bleeding can kill within minutes, and it is a leading cause of preventable death after injury, and a bystander can act on it. A short course teaches the skills in this entry, and a bystander can use them before help arrives.

WHAT IT IS. Hemorrhage is severe bleeding. External hemorrhage is the kind you can see and reach. The three techniques are direct pressure, wound packing and a tourniquet, and they are used in escalation: pressure for most wounds, packing for deep wounds where pressure alone is not enough, and a tourniquet for life-threatening bleeding from an arm or a leg that pressure does not control. When a limb is bleeding massively, as after an amputation, a tourniquet may be applied first, as your training directs.

WHY IT MATTERS. The adult body holds a limited amount of blood, and a heavy bleed can drain it fast. Bleeding that spurts, pools on the ground, soaks through clothing, or does not stop with firm pressure is life-threatening. The signs of shock from blood loss are those of Entry 3.5. The best moment to act is before they appear.

DIRECT PRESSURE. Press firmly on the wound with a clean cloth, gauze or a dressing, using both hands. Do not lift the dressing to look. If blood soaks through, add more material on top and keep pressing. Hold pressure until help arrives or until the bleeding stops. This is the first step for most bleeding.

WOUND PACKING. When the wound is deep and the bleeding does not stop under pressure, pack the wound. Push gauze, or gauze that contains a clotting agent if you have it, deeply into the wound and onto the bleeding source until the wound is filled, then

press firmly over it with both hands and keep pressing. Pressure over a clotting-agent dressing is held for at least 3 minutes, or as your protocol directs.

TOURNIQUETS. A tourniquet is a band that is tightened around an arm or leg above a wound to stop blood flow. It is for life-threatening bleeding from a limb that has not been controlled by other measures. It works, and it hurts. Apply it between the wound and the heart, about 2 to 3 inches (5 to 7.5 centimeters) above the edge of the wound and not over a joint. If the source of the bleeding cannot be found or it is unsafe to look, apply it as high on the limb as possible. Tighten it until the bleeding stops, and note the time. Do not loosen it; leave that to clinicians.

WOUNDS WHERE A TOURNIQUET DOES NOT FIT. A tourniquet works only on a limb. Bleeding from the groin, the armpit, the shoulder or the neck is too close to the trunk for an ordinary limb tourniquet, and pressure and packing are the tools there; trained clinicians may use a special device at the groin or armpit. Press hard and keep pressing, and call for help at once. A nosebleed is usually managed by sitting upright, leaning forward and pinching the soft part of the nose, but a nosebleed after a head injury needs medical help.

LIMB LOSS AND AMPUTATIONS. A partly or fully severed limb can bleed heavily. Control the bleeding first with firm pressure, and use a tourniquet if the bleeding is life-threatening and pressure does not control it. Wrap any severed part in clean gauze or cloth, place it in a plastic bag and keep the bag cool, never with the part itself directly on ice, and send it with the patient.

TOURNIQUET PITFALLS. A tourniquet that is too loose can make bleeding worse, because it blocks the veins that carry blood away while the arteries still push blood in, so tighten until the bleeding stops. A tourniquet is painful, and the patient may beg

you to remove it; explain why it stays on. If one tourniquet does not stop the bleeding, a second may be placed next to it, closer to the heart, as your training directs. Leave the tourniquet visible so that no one overlooks it.

Figure 13 shows direct pressure and packing; notice that the gauze fills the wound and the hands press over it.

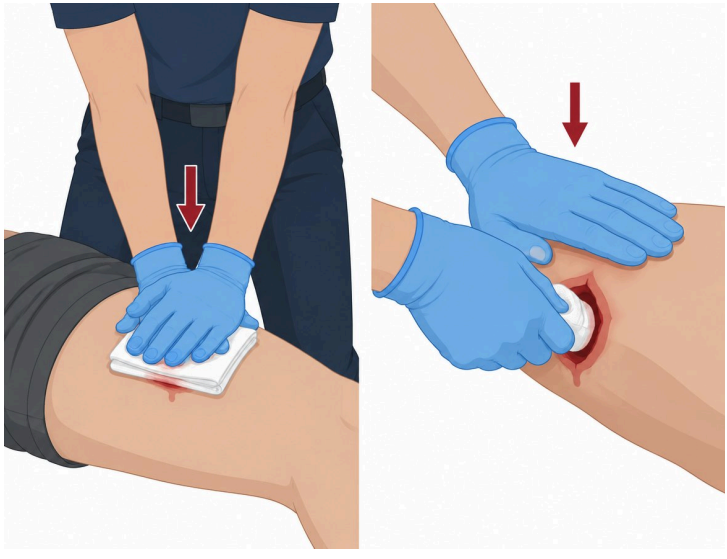


Figure 13. Direct pressure over a wound and gauze packed into a deep wound.

Figure 14 shows a tourniquet; notice that it sits between the wound and the heart, a few inches above the wound.



Figure 14. A tourniquet applied a few inches above a bleeding wound, between the wound and the heart.

PROCEDURE: CONTROL SEVERE BLEEDING

Purpose: to stop life-threatening bleeding while help comes. Scope: follows your training level and local protocol.

1. Make the scene safe, put on gloves, and call for help.
2. Find the bleeding and expose the wound.
3. Press firmly on the wound with a clean dressing or cloth, and keep pressing.
4. If the wound is deep and bleeding continues, pack it with gauze and press over the packing.

5. If bleeding from an arm or leg is severe and not controlled, apply a tourniquet between the wound and the heart, 2 to 3 inches (5 to 7.5 centimeters) above the wound and not over a joint, and tighten it until the bleeding stops.
6. Write down the time the tourniquet was applied and tell the responders.
7. Keep the patient lying down, warm and calm, and watch for signs of shock.
8. Do not lift the dressings or loosen the tourniquet; if blood soaks through, add more material on top.

> ⚠ **SAFETY**
>
> Wear gloves and avoid contact with blood. Do not remove an object that is stuck in a wound; pad around it and press around it. A tourniquet must not be placed over a joint. Do not use a tourniquet for bleeding from the neck, chest or abdomen.

STOP AND CALL FOR HELP IF bleeding does not slow, the patient becomes pale, confused or faint, or you cannot reach the wound.

The sequence follows what works. Pressure acts on the vessel directly and is available to everyone. Packing reaches a bleeding source that pressure on the surface cannot. A tourniquet stops flow to the whole limb, which is why it is held back for the bleeding that other methods have failed to control. Writing down the time matters because the receiving team must know how long the limb has been without blood.

RED FLAGS

Blood that spurts or pools, a dressing that soaks through, pale or cold skin, faintness or confusion, a fast weak pulse, or bleeding from the neck, groin or armpit. Call for help at once and keep pressing.

NURSING AND RESPONDER PRIORITIES. Control severe bleeding before the airway: press, pack deep wounds, use a tourniquet for limb bleeding that stays uncontrolled, and record the time.

See also: Entry 1.3 (primary survey); Entry 3.5 (shock); Entry 4.1 (trauma); Entry 6.1 (tourniquets in children).

4.3 Head, Spine and Chest Injuries

These injuries threaten the organs that control breathing and thinking, and they often occur together. Your task is to recognize them, to avoid making them worse, and to get help.

HEAD INJURIES. A blow to the head can cause a bruise, a concussion, bleeding inside the skull, or a skull fracture. The patient may have a headache, nausea, confusion, drowsiness, a change in the pupils, or a seizure. A patient who becomes more drowsy, more confused or who vomits repeatedly after a head injury is worsening and needs urgent help. Check the level of consciousness with the scale described in Entry 1.3 and recheck it, since the trend tells you most. Scalp wounds can bleed heavily and are controlled with firm pressure, but if the skull may be broken (a dent or soft area, visible bone fragments, or clear fluid from the nose or ears), do not press hard on the wound; cover it with a clean dressing and press only around it.

SPINAL INJURIES. Suspect a spinal injury after a high-risk mechanism, after a head injury, or when the patient has neck or back pain, numbness, weakness or tingling, is confused or intoxicated, or has another very painful injury that could hide neck pain. The aim is to limit movement of the spine until the patient is assessed. If the airway needs opening, use the jaw thrust, which moves the neck less than a head tilt; Entry 2.2 described it. Spinal motion restriction means keeping the head, neck and trunk in line and minimizing movement during care and transfer. It does not mean tying a patient rigid to a board in every case, and practice has changed in recent years, so follow your local protocol. In the first moments, a responder can hold the head gently in line with the body and ask the patient not to move.

FACE AND NECK INJURIES. Facial injuries can bleed heavily and bring blood and swelling into the airway. If the patient is awake and there is no spinal concern, let the patient take the position that lets blood drain from the mouth best, usually sitting up and leaning forward, and have suction ready if you are trained and your protocol allows. A wound to the front of the neck can bleed heavily and can injure the airway. Apply direct pressure with a clean dressing, enough to control the bleeding without squeezing the airway, and call for help.

HELMETS AND OTHER EQUIPMENT. A helmet protects the head and can hide a face or neck injury. Remove it only if your training and protocol direct, and if you must, do it with a second person who holds the head and neck in line. Do not remove a helmet that fits well if the patient is breathing without difficulty and help is coming. Leave in place any object that is stuck in a wound.

CONCUSSION. A concussion is a brain injury from a blow or a jolt. The patient may be dazed, forget what happened, have a headache, feel dizzy or nauseous, or be sensitive to light and noise.

A concussion can be present with no loss of consciousness. Any patient with a suspected concussion should be assessed by a clinician, should not return to sport that day or until a healthcare provider clears it, and should be watched for the warning signs in the Red Flags box.

CHEST INJURIES. Chest injuries include broken ribs, a bruised lung, a collapsed lung and bleeding into the chest. Recognize them by pain on breathing, fast shallow breathing, difficulty breathing, bruising, an abnormal movement of the chest wall, or the sound of air entering a wound. They need urgent help. This book teaches recognition only. Give oxygen as directed, let the patient find the position of greatest comfort, and cover an open chest wound as your protocol directs. Giving medicines is outside many roles; follow your scope and medical direction.

Figure 15 shows the chest wall and lungs; notice the red marks, which are the common sites of rib fracture.

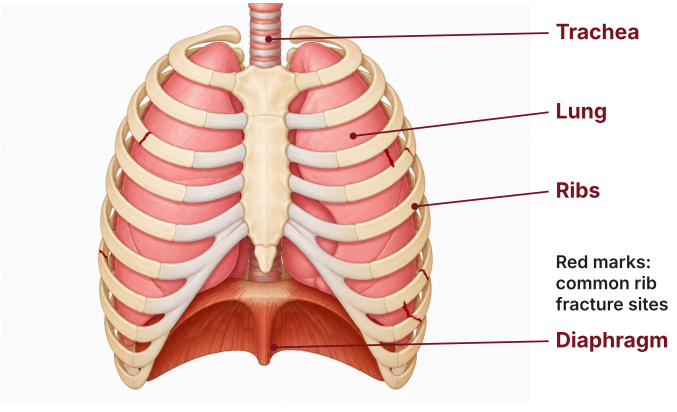


Figure 15. The chest wall and lungs, with the common sites of rib fracture marked in red.

RED FLAGS

Worsening drowsiness or confusion, repeated vomiting, a seizure, unequal pupils, clear fluid from the nose or ears, weakness or numbness in the limbs, severe neck pain, difficulty breathing, or a wound that sucks air. Call for help at once.

SAFETY

Do not move a patient with a possible spinal injury unless the scene is unsafe or your protocol directs. If you must move the patient, keep the head, neck and body in line. Do not remove a helmet unless your training and protocol direct it.

STOP AND CALL FOR HELP IF the airway or breathing is threatened, the patient cannot be kept still, or the scene becomes unsafe.

SCOPE NOTE

Spinal motion restriction techniques are taught and practiced with supervision. A student or lay responder keeps the patient still, holds the head gently in line if trained, and waits for the responders.

NURSING AND RESPONDER PRIORITIES. Suspect the spine after a high-risk mechanism, limit movement, open the airway with a jaw thrust, and watch the level of consciousness for change.

See also: Entry 2.2 (jaw thrust); Entry 4.1 (mechanism); Entry 6.1 (children and the spine).

4.4 Fractures, Dislocations and Splinting

A broken bone is painful and often obvious, but the injury that matters most may be to the nerves and blood vessels beside it. Splinting protects them.

WHAT IT IS. A fracture is a break in a bone. A dislocation is a joint forced out of its normal position. A sprain is a stretched or torn ligament. The three can look alike. Treat any painful, swollen or deformed limb after an injury as a possible fracture, since you cannot tell by looking.

CHECKING CIRCULATION, SENSATION AND MOVEMENT. These three checks are repeated before and after a splint. Circulation: feel for a pulse beyond the injury, press a fingertip or toe and watch the color return, the capillary refill check of Entry 1.3,, and compare the limb with the other side. Sensation: touch a fingertip or toe lightly and ask whether the patient can feel it. Movement: if it causes no pain, ask the patient to wiggle the fingers or toes beyond the injury; do not ask the patient to move the injured part itself. A limb that loses any of the three after splinting is a sign that the splint is too tight or that the injury has worsened.

RED FLAGS

A limb that is pale, blue, cold or without a pulse, an open wound with bone showing, a deformed thigh or pelvis, or signs of shock. Call for help at once.

PELVIS, HIP AND THIGH. Fractures of the pelvis and the thigh bone can cause heavy bleeding and are emergencies. A patient with a hip fracture often cannot stand, and the leg may look short-

ened and turned outward. Do not attempt to splint a pelvis yourself without training; keep the patient flat and still, treat for shock, and call for help.

DISLOCATIONS AND SPRAINS. A dislocated joint looks out of place and the patient guards it. Do not try to put it back. Support it in the position of comfort with padding and splint it as found. A sprain is treated the same way until a clinician has examined the limb. The familiar advice of rest, a cold pack wrapped in cloth, light compression and elevation fits a minor injury, and only a clinician can say that an injury is minor.

RECOGNITION. The signs include pain, swelling, bruising, deformity, an inability to use the limb, and sometimes a grating sensation. An open fracture has a wound with bone exposed or nearby, and it is more urgent. A limb that is pale, cold, blue or without a pulse beyond the injury is an emergency.

PROCEDURE: SPLINT A LIMB

Purpose: to keep a broken limb still and protect it from further injury. Scope: follows your training level and local protocol.

1. Make the scene safe and treat any severe bleeding first.
2. Check the circulation, sensation and movement below the injury and note what you find.
3. Cover any open wound with a clean dressing.
4. Support the limb in the position found, without forcing it straight.
5. Place a splint that holds the joint above and the joint below the injury still.
6. Pad the splint and secure it firmly without cutting off the circulation.

7. Check the circulation, sensation and movement again after splinting, and note the time.
8. Raise the limb if you can, apply a cold pack wrapped in cloth, and keep the patient comfortable.

Figure 16 shows a splinted limb; notice that the splint reaches past the joint above and the joint below the injury.



Figure 16. A splint that holds the joint above and the joint below a broken limb.

> **⚠ SAFETY**
>
> Do not try to push a bone back or straighten a badly deformed limb unless your protocol directs. Do not apply a cold pack directly to the skin. If the limb loses its pulse or becomes pale or blue, loosen the splint and call for help.

STOP AND CALL FOR HELP IF the limb is pale, blue, cold or pulseless, the bone is exposed, or the patient has signs of shock.

Holding the joint above and the joint below stops the broken ends from moving when the muscles pull, and padding spreads the pressure so that the splint does not press on skin and vessels. The checks before and after are the heart of the skill. A splint that is too tight can cut off blood flow and cause serious harm, and the only way to know that it has not is to compare what you found before and after.

NURSING AND RESPONDER PRIORITIES. Check circulation, sensation and movement before and after splinting the joint above and below, and call for help for a limb that is pale, cold or pulseless.

See also: Entry 4.1 (trauma); Entry 4.2 (bleeding); Entry 6.1 (children).

4.5 Burns

A burn is damage to the skin and the tissues beneath from heat, chemicals, electricity or radiation. The size and depth decide how serious it is, and burns to the face, hands, feet, genitals and joints are more serious than their size suggests.

WHAT IT IS. Burns are grouped by depth. A superficial burn reddens the skin and is painful. A partial-thickness burn blisters. A full-thickness burn may look white, leathery or charred and may be painless because the nerves are destroyed. A burn can deepen over hours.

ESTIMATING SIZE. The size is stated as a percentage of the total body surface area (TBSA). In adults, the rule of nines assigns about 9 percent to the head and neck, 9 percent to each arm, 18 percent to each leg, 18 percent to the front of the trunk, 18 percent to the back of the trunk and 1 percent to the genital area. Each of the head, arm and leg shares is split evenly between the front and the back. Count only burns that blister or are deeper; simple redness is not counted. A quick alternative is that the palm of the patient's hand, with the fingers, is about 1 percent of the body surface.

Figure 17 shows the body divided for the rule of nines; notice that the front and back views each carry half of the head, arm and leg shares.

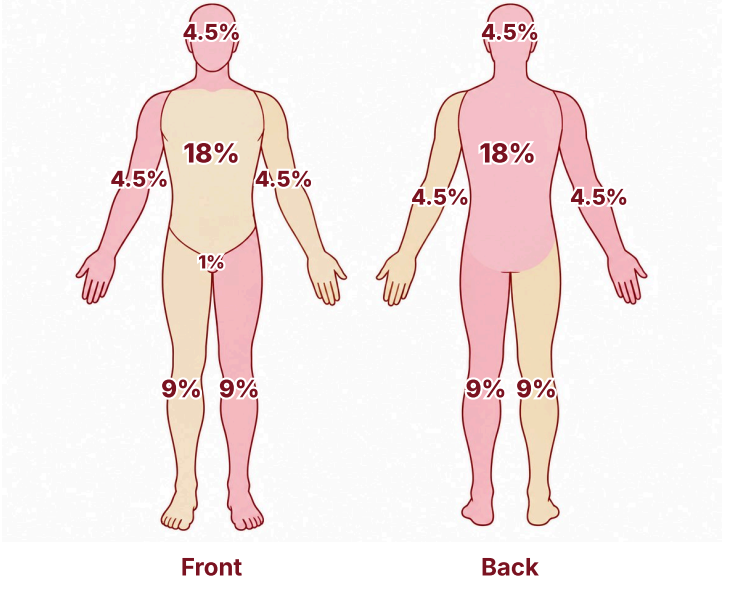


Figure 17. The adult body in front and back views, with each region's share of the body surface.

COOLING LARGE BURNS. Cooling reduces damage and eases pain, but a patient with a large burn can become dangerously cold. Cool for the time your protocol directs (for most burns from heat, at least 10 minutes and ideally 20 minutes, and not beyond 40 minutes), then cover the burn loosely and keep the rest of the body warm. Do not cool with ice or very cold water, and watch for signs of getting too cold, such as heavy shivering or confusion; if they appear, stop cooling and keep the patient warm, as your protocol directs. For a burn that covers a large area, call for help before cooling and do not let cooling delay transport.

CHEMICAL, ELECTRICAL AND SMOKE INJURIES. Chemical burns continue as long as the chemical is on the skin. Brush off any dry powder first, then flush with large amounts of running water, and take care that the runoff does not reach you or another person. An electrical injury can leave a small entry burn with serious damage beneath: do not touch the patient until the current is off, then call for help. Hoarseness, soot around the mouth or nose, or singed facial hair after smoke or flame points to an airway burn, which is an airway emergency.

CHILDREN AND OLDER ADULTS. Children have thinner skin, so the same exposure burns deeper, and the head is a larger share of their body surface, so a burn to the head covers a larger percentage. Older adults also have thin skin and often several illnesses. Burns in both groups need earlier help, and a burn in a child with an unclear history can be a sign of abuse, which Entry 6.6 covers.

FIRST PRIORITIES. Stop the burning process and make sure the scene is safe. Remove the patient from the source. Remove rings and tight clothing near the burn before swelling begins, unless stuck to the skin. Cool the burn with cool running water for at least 10 minutes and ideally 20 minutes, or as your protocol di-

rects. Cover it loosely with a clean, non-stick dressing. Keep the patient warm elsewhere, since large burns cause heat loss. Check the airway if the burn is on the face or was caused by smoke or flame in a closed space, because swelling can close it. Do not apply creams, ice or butter, and do not break blisters.

RED FLAGS

Burns to the face, hands, feet, genitals or joints, burns that go all the way around a limb, electrical or chemical burns, burns with smoke inhalation, hoarseness or soot around the mouth, and burns in children and older adults. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Stop the burning, cool the burn as directed, cover it loosely, protect the airway, keep the patient warm, and call for help for any significant burn.

See also: Entry 2.1 (airway); Entry 6.1 (burns in children).

4.6 Heat and Cold Emergencies

The body works only in a narrow temperature range. Heat or cold can push it past that range, and the most severe forms are emergencies.

HEAT ILLNESS. Heat exhaustion causes heavy sweating, weakness, headache, nausea and a fast pulse. Heat stroke is the emergency form. The body temperature rises to a dangerous level, and the patient becomes confused, faints, has a seizure or collapses. Move the patient to a cool place, remove excess clothing, call for help, and cool the body quickly with whatever method your protocol directs, such as cool water or wet cloths with fanning. Give cool drinks only to a patient who is fully alert.

COLD ILLNESS. Hypothermia is a fall in the body temperature. Early signs are shivering, clumsiness and confusion. As it worsens, shivering stops, the patient becomes drowsy and the pulse slows. Move the patient from the cold, handle them gently, replace wet clothing with dry, insulate from the ground and the air, and call for help. Do not rub the limbs or give alcohol.

WHO IS AT RISK. Infants, older adults, people who work or exercise outdoors, people with long-term illness, and people taking some medicines are at higher risk of heat illness. Cold illness is more likely in people who are wet, tired, hungry, intoxicated, or ill, and it can occur at temperatures that are above freezing. Ask how long the patient has been exposed and what they were wearing.

COOLING METHODS. In heat stroke, speed matters more than the method. Immersion in cold water cools fastest where it is available and safe, and ice packs in the armpits, neck and groin or wet cloths with fanning are alternatives. Keep the airway and consciousness under watch during cooling, and stop when the patient is cool enough as your protocol directs, since overcooling is also possible.

FROSTBITE. In frostbite, the tissue itself freezes, most often in the fingers, toes, ears and nose. The skin may look white or waxy and feel numb. Move the patient to shelter, remove wet or tight clothing and rings, and cover the area loosely with dry material. Do not rub the area, and do not rewarm it if there is a chance it could freeze again before the patient reaches care.

Figure 18 shows how heat and cold illness progress; notice that the signs and the first actions change at each stage.

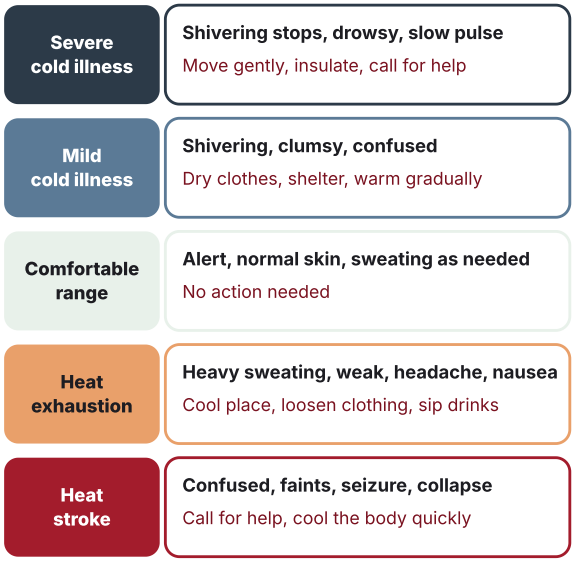


Figure 18. A continuum from cold illness to heat illness with the signs at each stage.

RED FLAGS

Confusion, fainting, a seizure or collapse in the heat, or shivering that stops with drowsiness or a slow pulse in the cold. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Recognize confusion in a hot or cold setting as an emergency, cool or warm the patient as directed, handle gently, and call for help.

See also: Entry 6.4 (older adults).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Tomas, a paramedic student, was cycling on a trail when he came across a rider who had fallen on a rocky section. The rider, Ellie, sat beside her bike, holding her thigh. There was blood running through her fingers onto the ground. Tomas stopped, looked up and down the trail, and put on the gloves from his bag. He put his phone on speaker beside him, called the emergency number and gave the location.

He pressed a folded spare shirt from his pack firmly against the wound with both hands. Blood soaked through, so he added more on top and kept pressing. It did not slow. He packed the wound with gauze from his kit and pressed hard over it, but the blood kept coming. Ellie went pale and said she felt cold. Tomas reached for the tourniquet in his kit, applied it on the thigh a few inches above the wound, between the wound and her body, and tightened it until the bleeding stopped. He wrote the time on her arm with a pen. When Ellie winced and tried to reach for the band, he told her it would hurt but was holding the bleeding back, and he left the dressing in place. He kept her lying down, covered her with his jacket, and watched her color and her thinking while he waited. He had checked that the band sat above the wound and not over the knee. When the paramedics arrived, he told them what he had seen on the trail, the time on her arm and that he had not loosened the band.

Care Priorities

CARE PRIORITIES

When injuries compete, rank them by how soon each can kill. Bleeding that you can stop goes first, once the scene is safe and you can reach the patient. Airway and breathing follow, with the spine protected while you work, because a spinal injury is something you guard against and do not clear yourself. A dramatic wound is not always the quickest threat, so look past it for the quiet signs of shock before you settle on a plan. Cold and wet make every injury worse, so shelter the patient from the weather from the first minute. At handover, describe what you saw at the scene as well as what you found on the patient, because the mechanism tells the team where to look for injuries you could not see.

Review Questions

1. In a trauma patient, which problem is controlled before the airway, as most protocols direct?
   - **A)** A deformed wrist.
   - **B)** Severe external bleeding.
   - **C)** A bump on the head with a headache.
   - **D)** A superficial burn on a finger.

2. Which is a high-risk mechanism of injury?
   - **A)** A short trip on level ground.
   - **B)** A cut finger from a kitchen knife.
   - **C)** A twisted ankle while walking.
   - **D)** A fall from a height.

3 What is the first technique for controlling bleeding from a wound?

- **A)** Apply direct pressure with a clean dressing.
- **B)** Apply a tourniquet.
- **C)** Wash the wound.
- **D)** Lift the dressing often to check.

4 Where is a tourniquet applied on a bleeding limb?

- **A)** Directly over the wound.
- **B)** Over a joint.
- **C)** Between the wound and the heart, a few inches above the wound.
- **D)** Below the wound.

5 What is the aim of spinal motion restriction?

- **A)** To straighten the neck so that injuries can be seen.
- **B)** To move the patient to the ambulance as fast as possible.
- **C)** To keep every patient lying flat on a board.
- **D)** To keep the head, neck and trunk in line and limit movement.

6 When splinting, what should you check before and after?

- **A)** How far the limb can be straightened.
- **B)** Circulation, sensation and movement below the injury.
- **C)** Whether the patient can walk on the limb.
- **D)** The patient's age.

7 About what share of the body surface is the palm of the patient's hand, with fingers?

- **A)** 9 percent.
- **B)** 5 percent.
- **C)** 1 percent.
- **D)** 18 percent.

8 Short case. On a hot afternoon a worker who has been digging outdoors becomes confused and then faints. His skin is hot and flushed. List, in order, the first four things you would do and say why this is an emergency.